



PROJECT REPORT



AWS CLOUDWATCH ALARM & AMAZON SNS NOTIFICATION MANAGEMENT

Monitoring AWS EC2 with Real-Time Alerts

Monitor EC2 instance performance using AWS CloudWatch and send automated notifications using Amazon SNS



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OBJECTIVE
Automate Monitoring & Alerts
for Better System Reliability



TOOLS & SERVICES USED
AWS EC2, CloudWatch,
SNS, IAM

Project Introduction

Imagine you own a large shopping mall. Since you can't watch every corner of the mall all the time, you hire a security guard to keep an eye on everything.

As long as everything is normal,

But if there is a fire, or any emergency, the guard immediately calls the manager so that quick action can be taken before the situation becomes serious.

Now think about a computer server in the same way.

A server also needs someone to monitor its health 24/7.

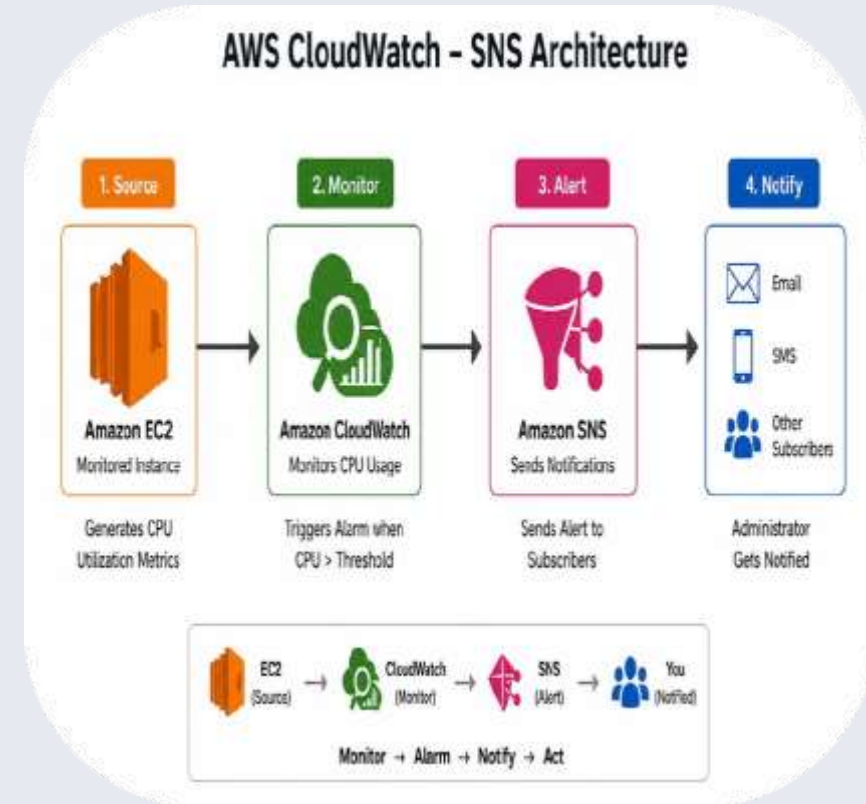
If its CPU usage becomes too high and nobody notices, the application may become slow or even stop working.

That's exactly the problem my project solves.

In my project :

- Amazon EC2 - is the server where the application is running.
- Amazon CloudWatch - acts like the security guard. It continuously monitors the server's CPU usage. If the CPU usage crosses the defined threshold, CloudWatch immediately triggers an alarm.
- Amazon SNS - acts like the phone call. It instantly sends an email notification to the administrator.

The administrator receives the alert and takes quick action before the problem affects users. This is how my project provides real-time monitoring and automated notifications, helping improve system reliability and reduce downtime.



Why Monitoring Is Important ?

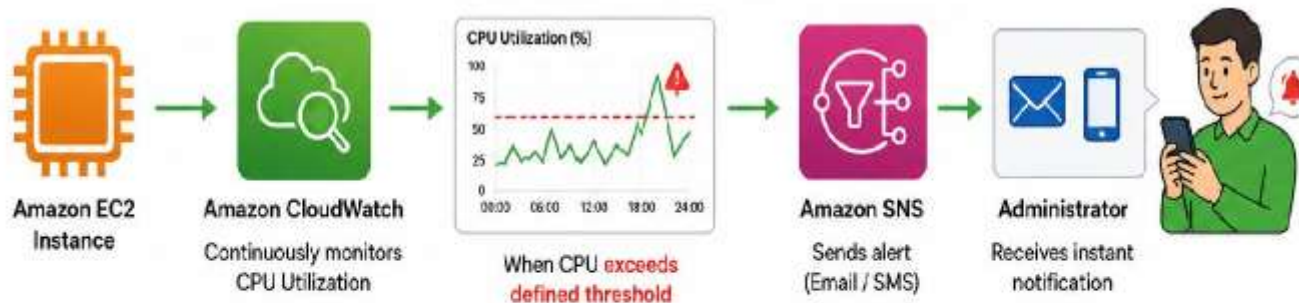
THE PROBLEM



Problem Statement

1. High CPU usage can reduce application performance
2. Manual server monitoring is time-consuming
3. Delayed alerts
4. Downtime affects users and business operations.

THE SOLUTION



Solution Statement

1. Deploy the application on an **Amazon EC2** instance.
2. Use **Amazon CloudWatch** to continuously monitor CPU utilization.
3. Configure a **CloudWatch Alarm** with a predefined CPU threshold.
4. Integrate the alarm with **Amazon SNS**.
5. Automatically send **email notifications**



Quick alerts enable quick action and prevent downtime before it affects users.

Which AWS Services Used In My Project

➤ This project uses three core AWS services.

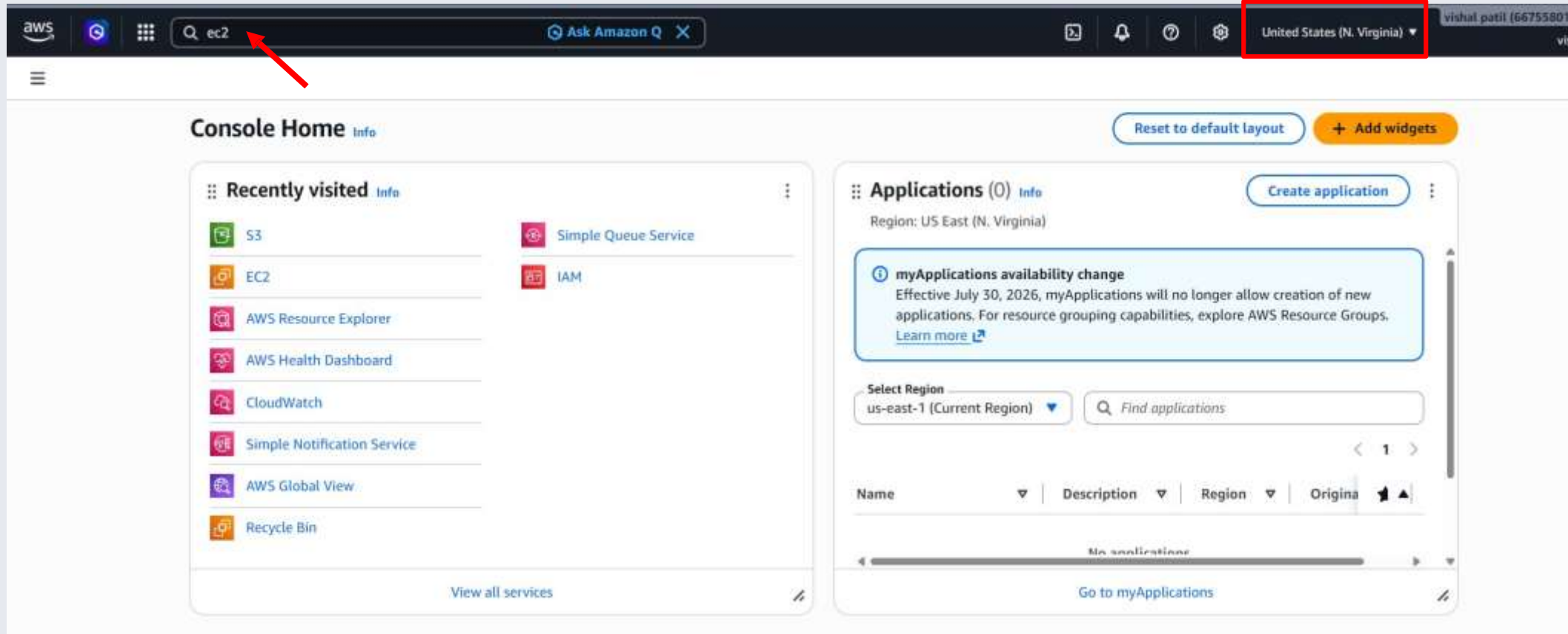
1. **Amazon EC2** (Elastic Compute Cloud)
2. **Amazon CloudWatch**
3. **Amazon SNS** (Simple Notification Service)



AWS Service	What It Does
Amazon EC2	Runs the server/application
Amazon CloudWatch	Monitors CPU usage
Amazon SNS	Sends email alerts

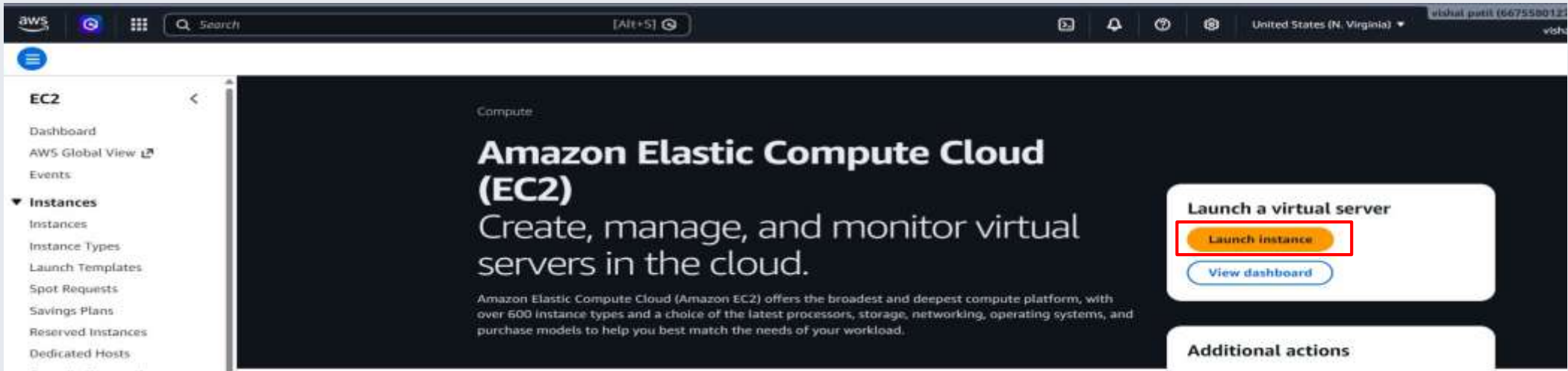
Implementation Steps

- ❑ Open AWS Console Management and Search EC2



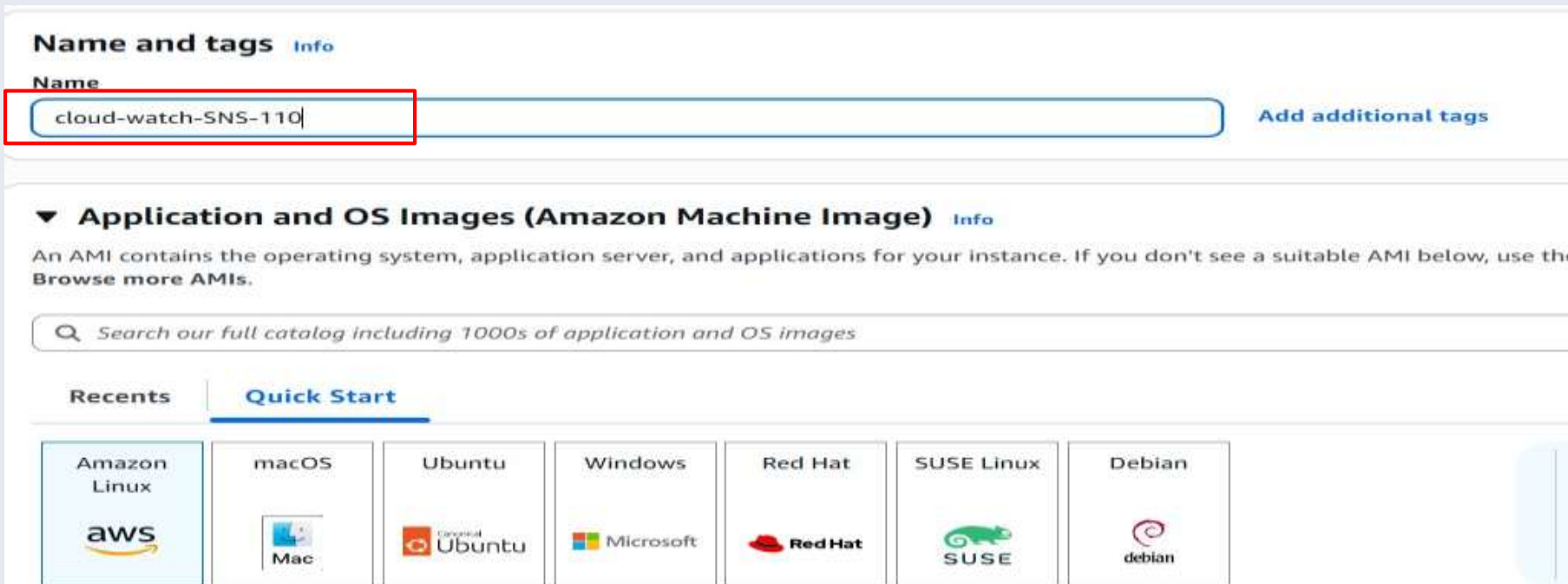
1. Log in to AWS
2. Select the region
3. Search EC2

❑ Launch Instance



The screenshot shows the AWS Management Console for the EC2 service. The left sidebar contains a navigation menu with options like Dashboard, AWS Global View, Events, and Instances. The main content area features the heading 'Amazon Elastic Compute Cloud (EC2)' and a subheading 'Create, manage, and monitor virtual servers in the cloud.' Below this, there is a description of the service. On the right side, there is a 'Launch a virtual server' section with a prominent orange 'Launch instance' button, which is highlighted with a red rectangular box. Below it is a 'View dashboard' button. Further down, there is an 'Additional actions' section.

❑ Give the name of Instance



The screenshot shows the 'Name and tags' section of the AWS Management Console. The 'Name' field is highlighted with a red rectangular box and contains the text 'cloud-watch-SNS-110'. To the right of the field is a link 'Add additional tags'. Below this section, there is a 'Application and OS Images (Amazon Machine Image)' section. It includes a search bar with the placeholder text 'Search our full catalog including 1000s of application and OS images'. Below the search bar, there are two tabs: 'Recents' and 'Quick Start'. Under the 'Quick Start' tab, there is a grid of seven AMI cards: Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, and Debian. Each card displays the respective logo and name.

- **Instance Type** – select t3 micro for free tier
- **Key pair** – proceed without key pair

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Proceed without a key pair (Not recommended)

Default value ▼

[↻ Create new key pair](#)

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

[Learn more](#)

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

[Create a new RDS database](#)

[Learn more](#)

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

Manage detailed monitoring

Create Load Balancer

Create a application, network gateway or classic Elastic Load Balancer

Create Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Create AWS budget

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

View all instances

- After create instance then click on view all instance

Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...
cloud-watch-S...	i-0dbba5d2558e8c59e	Running	t3.micro	Initializing	us-east-1d	ec2-54-211-253-67.co...	54.211.253.67

Launch instances

Launch instance from template

Migrate a server

Connect

Stop instance

Start instance

Reboot instance

Hibernate instance

Terminate (delete) instance

Instance diagnostics

Instance settings

Networking

Security

Image and templates

Storage

Monitor and troubleshoot

Manage detailed monitoring

Manage CloudWatch alarms

Configure CloudWatch agent

Get system log

Get instance screenshot

Connect to serial port

Replace root volume

Visit SSM Fleet Manager

i-0dbba5d2558e8c59e (cloud-watch-S

Details

Status and alarms

Mo

Instance summary Info

Instance ID

i-0dbba5d2558e8c59e

IPv6 address

Private IPv4 addresses

172.31.41.130

Public DNS

- Right click on instance
- Click on monitor and troubleshoot
- Select manage cloudwatch alarms

Alarm notification [Info](#)

Configure the alarm to send notifications to an Amazon SNS topic when it is triggered.

cloud-watch-SNS

- Set cloudwatch name

Alarm thresholds

Specify the metric thresholds for the alarm.

Group samples by

Average

Alarm when

>=

Consecutive period

1

Alarm name

awsec2-i-0dbba5d2558e8c59e-GreaterThanOrEqualToThreshold-CPUUtilization

Alarm description

Alarm on instance i-0dbba5d2558e8c59e: Triggered when CPUUtilization >= 0.99 for 1 consecutive 5-minute periods.

Type of data to sample

CPU utilization

Percent

40

- Set percentage

Period

5 Minutes

Sample metric data [Info](#)

Sample metric data for i-0dbba5d2558e8c59e

☐ Alarm recommendations [?](#)

[Investigate with AI - new](#)

1h

3h

12h

1d

3d

1w

Custom

UTC timezone



[Explore related](#)



CPU utilization (Average)

Percent

40% 40

20%

0%

13:40

13:50

14:00

14:10

14:20

14:30

14:40

14:50

15:00

15:10

15:20

15:30

15:40

15:50

16:00

16:10

16:20

16:30

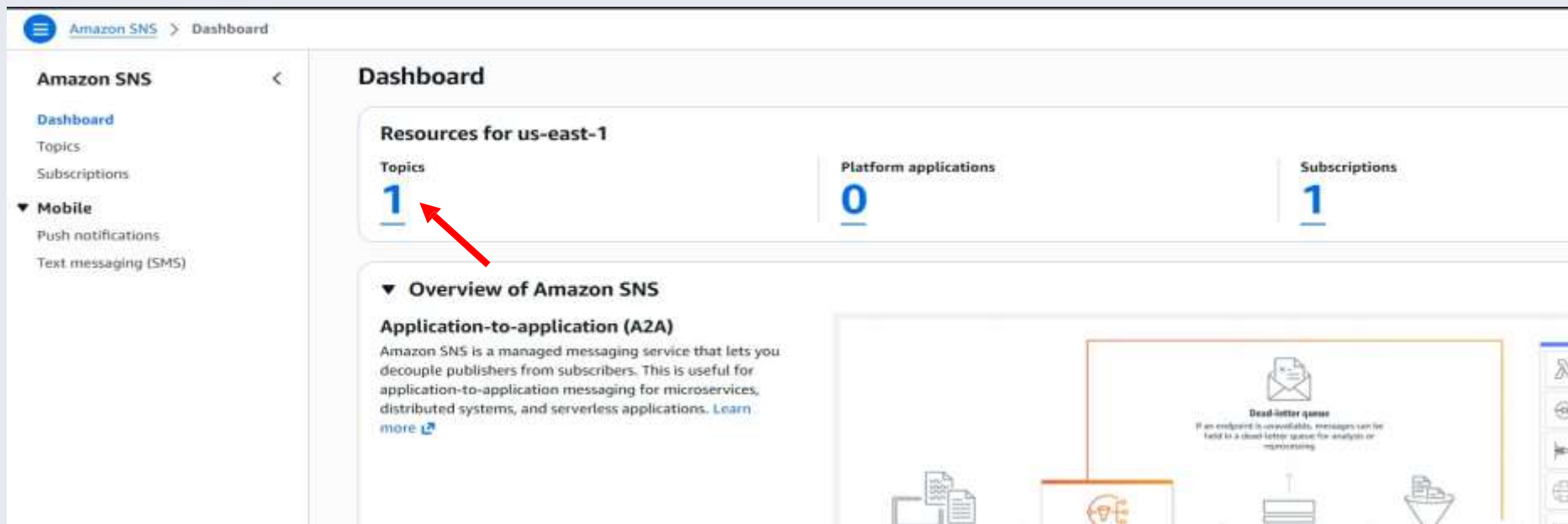
CPUUtilization

Cancel

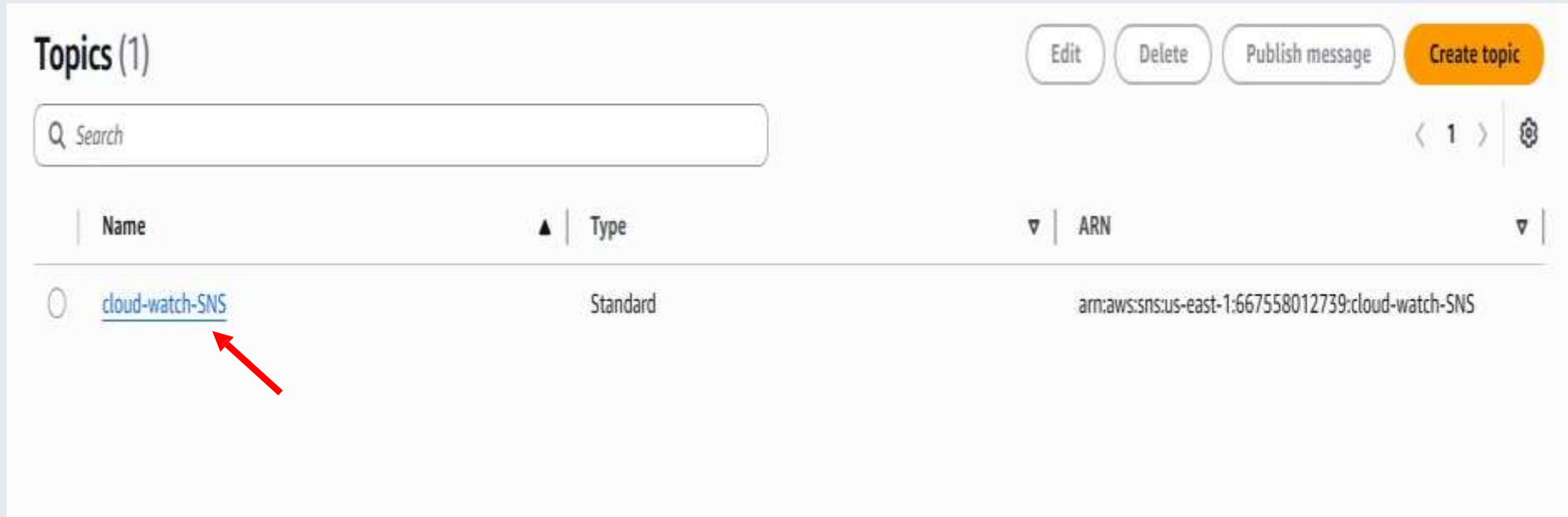
Create

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar with 'SNS' entered, and a 'Ask Amazon Q' button. The left sidebar shows the navigation menu with 'EC2' selected. The main content area displays 'Services' and 'Features' for SNS, including Simple Notification Service, AWS Sustainability, and Route 53 Resolver. The 'Instances' tab is selected in the left sidebar.

- Back to EC2 instance Tab and search SNS



- After that interface click on Topics



- Click on cloud-watch-SNS

Amazon SNS > Subscriptions

Amazon SNS

Dashboard

Topics

Subscriptions

▼ Mobile

Push notifications

Text messaging (SMS)

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Subscriptions (0) Edit Delete Request confirmation Confirm subscription Create subscription

Search

ID	Endpoint	Status	Protocol	Topic
No subscriptions found				

Create subscription

- Click on Create Subscription Button

Details

Topic ARN

Protocol
The type of endpoint to subscribe

Endpoint

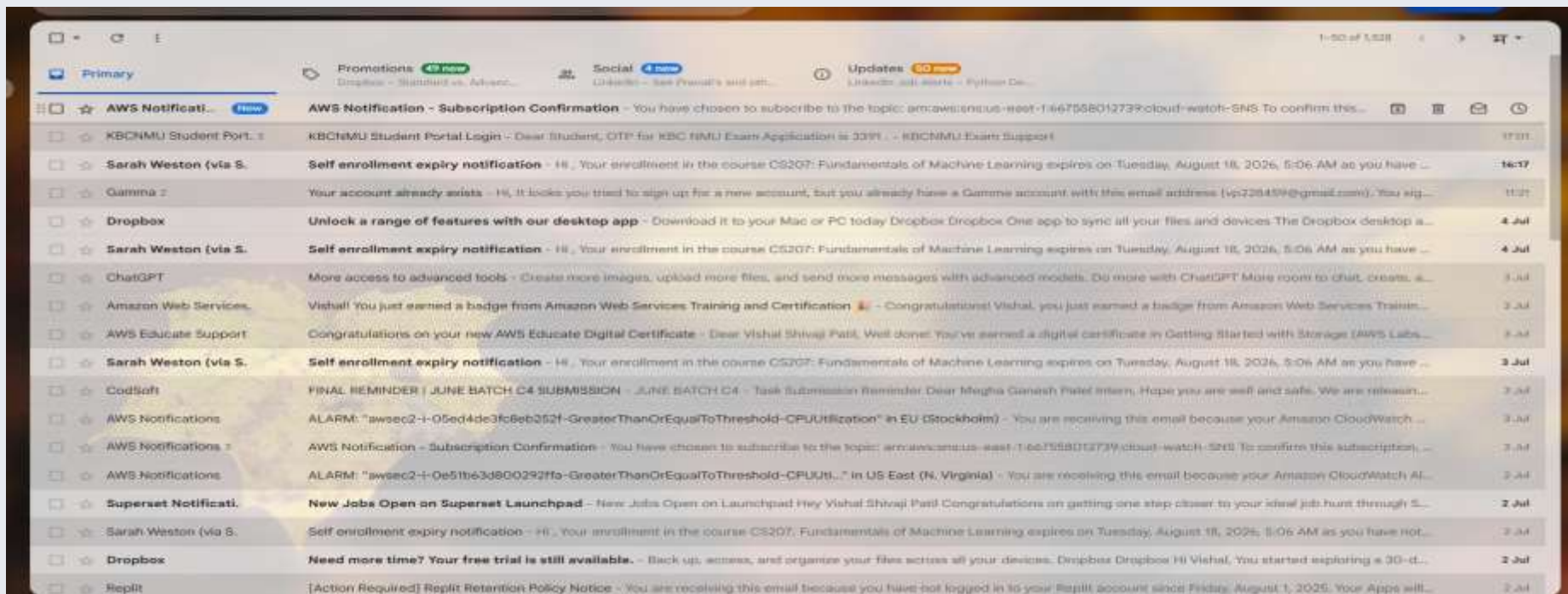
After your subscription is created, you must confirm it. [Info](#)

► **Subscription filter policy - optional** [Info](#)
This policy filters the messages that a subscriber receives.

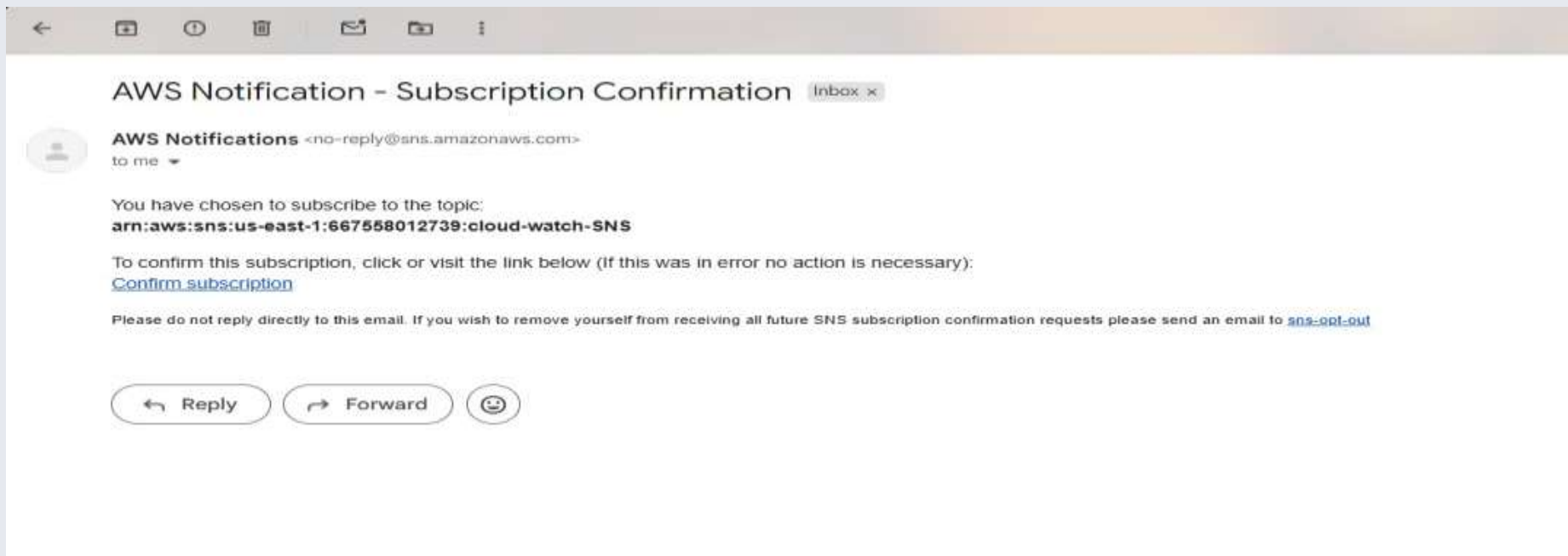
► **Redrive policy (dead-letter queue) - optional** [Info](#)
Send undeliverable messages to a dead-letter queue.

Cancel Create subscription

- Select protocol as a Email
- Put email on Endpoint
- Create Subscription



- Open Gmail
- Click on AWS notification



- Click on confirm subscription

Subscription: c6170bd7-17f0-464c-97f0-2846be054234

Edit

Delete

Details

ARN

 arn:aws:sns:us-east-1:667558012739:cloud-watch-SNS:c6170bd7-17f0-464c-97f0-2846be054234

Endpoint

vp228459@gmail.com

Topic

cloud-watch-SNS

Status

 Confirmed 

Protocol

EMAIL

Created CloudWatch alarm: awsec2-i-0425e18936d3af5ef-GreaterThanOrEqualToThreshold-CPUUtilization

Instances (1/1) [Info](#)

Last updated
1 minute ago

Refresh

Connect

Instance state ▾

Actions ▾

Launch instances ▾

Saved filter sets

Choose filter set ▾

Find Instance by attribute or tag (case-sensitive)

< 1 > ⚙

☒	Name 🔗 ▾	Instance ID	Instance state ▾	Instance type ▾	Status check	Availability Zone ▾	Public IPv4 DNS ▾	Public IPv4 ... ▾
☒	cloud-watch-s...	i-0425e18936d3af5ef	☒ Running 🔍 🔍	t3.micro	☒ 3/3 checks passed	us-east-1c	ec2-100-26-61-11.com...	100.26.61.11

Back to EC2 instance and connect the instance

By Mohit Gadilohar

Instances (1/1) [Info](#)

[Connect](#)[Instance state](#) ▼[Actions](#) ▼[Launch instances](#) ▼

Saved filter sets

[Choose filter set](#) ▼

< 1 >



☒	Name ✎ ▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability Zone ▼	Public IPv4 DNS ▼	Public IPv4 ... ▼
☒	cloud-watch-S...	i-0dbba5d2558e8c59e	Running 🔍 🔍	t3.micro	3/3 checks passed	us-east-1d	ec2-54-211-253-67.co...	54.211.253.67

i-0dbba5d2558e8c59e (cloud-watch-SNS-110)

[Details](#)[Status and alarms](#)[Monitoring](#)[Security](#)[Networking](#)[Storage](#)[Tags](#)

☒ Include metrics in the CWAgent namespace [?](#)

[Learn more](#) [🔗](#)

☐ Alarm recommendations [?](#)

[Investigate with AI - new](#) [?](#)

[1h](#)[3h](#)[12h](#)[1d](#)[3d](#)[1w](#)[Custom](#) [📅](#)[UTC timezone](#) ▼[Explore related](#) [?](#)

CPU utilization (%) [?](#) [⋮](#)

Percent

4.00%

Network in (bytes) [?](#) [⋮](#)

Bytes

4.40K

Network out (bytes) [?](#) [⋮](#)

Bytes

4.40K

Network packets in (count) [?](#) [⋮](#)

Count

4.4.4

CPU utilization (%)

5 minutes ▼

Average ▼

1h

3h

12h

1d

3d

1w

Custom 

UTC timezone ▼



Percent

50.9%

40.1

38.2%

25.5%

12.7%

0%

15:50 15:55 16:00 07-05 16:05 16:10 16:15 16:20 16:25 16:30 16:35 16:40 16:45 16:50

i-0dbba5d2558e8c59e (cloud-watch-SNS-110)

[View in metrics](#) 

[Close](#)

When CPU Utilization graph cross the 40% then check email notification

❑ Result

The screenshot shows a Gmail interface with a dark sidebar on the left. The main content area displays an email from 'AWS Notifications' with the subject 'ALARM: "awsec2-i-0dbba5d2558e8c59e-GreaterThanOrEqualToThreshold-CPUUtili..." in US East (N. Virginia)'. The email body contains the following information:

Alarm Details:

- Name: awsec2-i-0dbba5d2558e8c59e-GreaterThanOrEqualToThreshold-CPUUtilization
- Description: Alarm on instance i-0dbba5d2558e8c59e: Triggered when CPUUtilization >= 40 for 1 consecutive 5-minute periods.
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 datapoint [50.93600068817777 (05/07/26 16:44:00)] was greater than or equal to the threshold (40.0) " at "Sunday 05 July, 2026 16:49:38 UTC".
- Timestamp: Sunday 05 July, 2026 16:49:38 UTC
- AWS Account: 667556012739
- Alarm Arn: arn:aws:cloudwatch:us-east-1:667556012739:alarm:awsec2-i-0dbba5d2558e8c59e-GreaterThanOrEqualToThreshold-CPUUtilization

Threshold:

- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 40.0 for at least 1 of the last 1 period(s) of 300 seconds.

Monitored Metric:

- MetricNamespace: AWS/EC2
- MetricName: CPUUtilization
- Dimensions: [InstanceId = i-0dbba5d2558e8c59e]
- Period: 300 seconds
- Statistic: Average
- Unit: not specified

State Change Actions:

- OK:

At the bottom of the email, there are buttons for 'Reply', 'Forward', and a smiley face icon.